MALARIA SURVEILLANCE DASHBOARD

Nigeria, 2020–2023 — Capstone Documentation

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Tool: Microsoft Power BI Desktop

Reporting Period: 2020–2023 (2024 partial data excluded)

1. Project Overview

This capstone project analyzes four years (2020–2023) of quarterly malaria surveillance data across all 36 Nigerian states and the Federal Capital Territory. The dataset contains 600 rows in total, of which 592 represent complete quarterly reporting (37 states × 4 years × 4 quarters) and 8 rows represent a partial, incomplete 2024 Q1 drop covering only 8 South-South/South-East states.

The objective was to clean the dataset, build a two-page interactive Power BI dashboard, and extract actionable public-health insights covering disease burden, seasonal patterns, intervention coverage, vulnerable populations, and data quality.

This document follows the order the work was actually done: data cleaning first, then DAX measure development, then the resulting dashboard, then findings and recommendations.

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2. Data Cleaning & Preparation Process

The following cleaning and validation steps were carried out in Power Query before any DAX measures or visuals were built:

Step	Action Taken	Result
Completeness check	Checked all 14 original columns for missing values	0 missing values found
Duplicate check	Removed duplicate rows across all columns	0 duplicate rows found
Text standardization	Trimmed whitespace; cleaned invisible characters on State, Geopolitical_Zone, Quarter, Quarter_Months	Consistent text formatting across all categorical fields
State naming	Renamed "Federal Capital Territory (FCT)" to "FCT (Abuja)"	Improved chart/map label readability
Data type correction	Set Year/Population/Cases/Deaths to Whole Number; rate/percentage fields to Decimal Number	Correct type handling for aggregation and sorting
Quarter sort order	Added Quarter_Num (1–4) custom column so Q1–Q4 sort chronologically, not alphabetically	Correct chronological ordering in all quarter-based visuals
Geopolitical zone correction	Merged FCT back into North Central (Nigeria has 6 official geopolitical zones, not 7); applied via Table.ReplaceValue in Power Query	Zone classification matches Nigeria's standard administrative structure; confirmed 6 distinct values, 0% errors, 0% empty
Logical validation	Verified Malaria_Deaths never exceeds Reported_Malaria_Cases; verified all percentage fields fall within 0–100%	Passed — no violations found
2024 completeness check	Identified that 2024 contains only 8 of 37 states and only Q1	2024 excluded from all trend, KPI, and summary visuals to avoid understating totals
Burden tier classification	Added Burden_Tier column (Low / Moderate / High / Very High) based on Incidence_Rate_per_1000 thresholds (<20 / 20–35 / 35–50 / 50+)	Enables severity-based filtering and counts

Aggregation-level correction (identified during DAX/dashboard build, addressed retroactively as part of the cleaning process):

Several KPI cards initially used SUM instead of AVERAGE on percentage and rate fields (Incidence_Rate_per_1000, ITN_Coverage_Pct, Health_Facility_Reporting_Rate_Pct, Pregnant_Women_Cases_Pct). Because these columns repeat per state-quarter, summing them produced meaningless inflated figures (e.g. "3bn" population, "6K" incidence rate). All affected measures were corrected to use AVERAGE, or a filtered CALCULATE for point-in-time snapshots (e.g. Total Population using only the latest quarter to avoid 16x duplication across the reporting period).

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3. DAX Measures

Seven core DAX measures were built on top of the cleaned data, following one governing rule throughout: percentage and rate fields use AVERAGE, count fields use SUM, and any field affected by row repetition across quarters uses a filtered CALCULATE to prevent duplication.

Total Cases

```
Total Cases = SUM(Malaria_Data[Reported_Malaria_Cases])
```

Total Population

Filtered to a single snapshot period to avoid summing the same state's population across all 16 repeated quarterly rows. This correction fixed an earlier version of the measure that displayed an erroneous 3 billion.

```
Total Population =  CALCULATE(     SUM(Malaria_Data[Estimated_Population]),
FILTER(         Malaria_Data,         Malaria_Data[Year] = 2023 &&
Malaria_Data[Quarter] = "Q4"     ) )
```

Cases per 100k

```
Cases per 100k =  DIVIDE(     SUM(Malaria_Data[Reported_Malaria_Cases]),
SUM(Malaria_Data[Estimated_Population]),     0 ) * 100000
```

Avg Incident Rate

```
Avg Incident Rate = AVERAGE(Malaria_Data[Incidence_Rate_per_1000])
```

Avg ITN Coverage %

```
Avg ITN Coverage % = AVERAGE(Malaria_Data[ITN_Coverage_Pct])
```

Under-5 Cases Share

```
Under-5 Cases Share = AVERAGE(Malaria_Data[Under5_Cases_Pct])
```

Cases Fatality Rate %

```
Cases Fatality Rate % =  DIVIDE(     SUM(Malaria_Data[Malaria_Deaths]),
SUM(Malaria_Data[Reported_Malaria_Cases]),     0 ) * 100
```

KPI values these measures produce, as displayed on the dashboard (Section 4):

Measure	Displayed Value	Dashboard Page
Total Cases / Total Reported Cases (2020–2023)	145M	Page 1
Total Population	214M	Page 2
Cases per 100k	4.19K	Page 2
Avg Incident Rate	42.77	Page 2
Avg ITN Coverage %	59.30%	Page 1
Under-5 Cases Share	37.82%	Page 1
Cases Fatality Rate % ("Cases Report Ratio %")	0.27%	Page 1

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4. Resulting Dashboard

With the data cleaned (Section 2) and the DAX measures built (Section 3), the following two-page dashboard was produced in Power BI Desktop.

Page 1 — National Overview

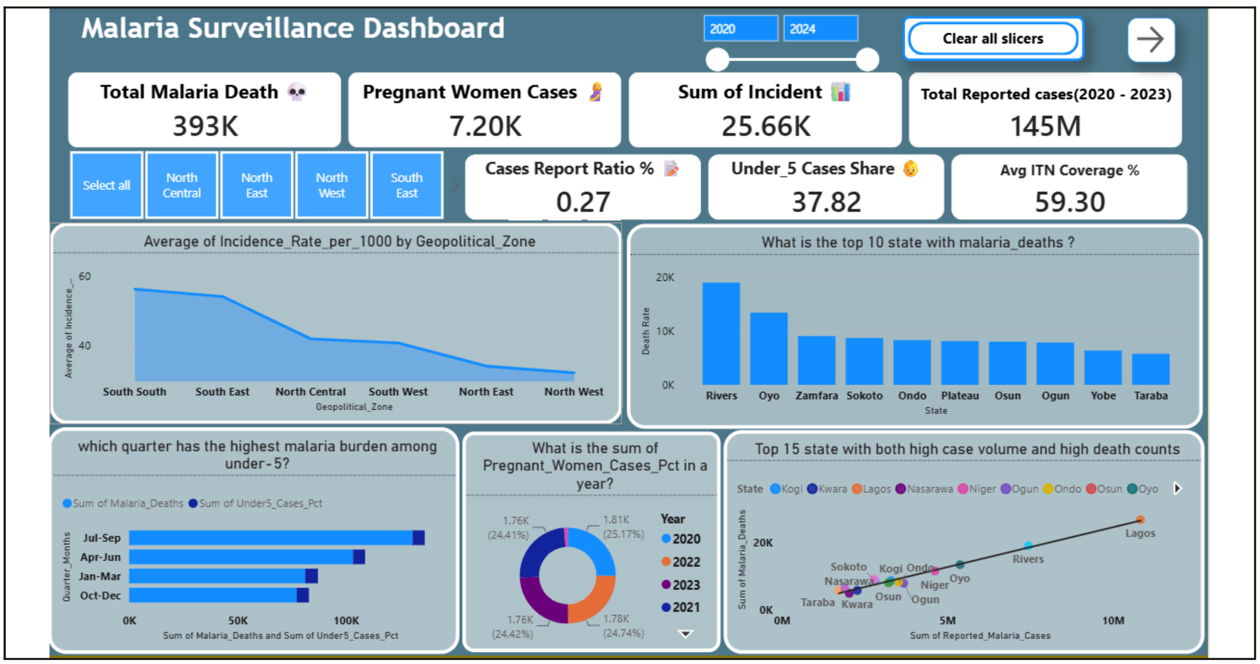

Figure 1. National Overview — headline KPIs, incidence by zone, top death-count states, seasonal under-5 burden, pregnant-women case share by year, and the 15-state case-volume-vs-death-count scatter.

- Header: "Malaria Surveillance Dashboard" title bar with a Year range slider (2020–2024) and a "Clear all slicers" reset button.
- KPI Row 1: Total Malaria Death (💀 393K) · Pregnant Women Cases (🤰 7.20K) · Sum of Incident (📊 25.66K) · Total Reported Cases 2020–2023 (📋 145M).
- Geopolitical_Zone tile slicer: Select all · North Central · North East · North West · South East · South South · South West.
- KPI Row 2: Cases Report Ratio % / Case Fatality Rate (✅ 0.27) · Under-5 Cases Share (👶 37.82) · Avg ITN Coverage % (🛏️ 59.30).

- "Average of Incidence_Rate_per_1000 by Geopolitical_Zone" — area chart; South South highest (~58), falling to North West lowest (~30).
- "What is the top 10 state with malaria_deaths?" — column chart; Rivers leads, followed by Oyo, Zamfara, Sokoto, Ondo, Plateau, Osun, Ogun, Yobe, Taraba.
- "which quarter has the highest malaria burden among under-5?" — horizontal bar by Quarter_Months (Jul-Sep, Apr-Jun, Jan-Mar, Oct-Dec); Jul-Sep highest.
- "What is the sum of Pregnant_Women_Cases_Pct in a year?" — donut chart by Year (2020–2023); shares stay tight, 24.4%–25.2%.
- "Top 15 state with both high case volume and high death counts" — scatter chart, Reported_Malaria_Cases (x) vs Malaria_Deaths (y); Lagos furthest along the trend line, followed by Rivers, Oyo, Ondo, Kogi, Niger, Sokoto.

Page 2 — Geographic & Data Quality

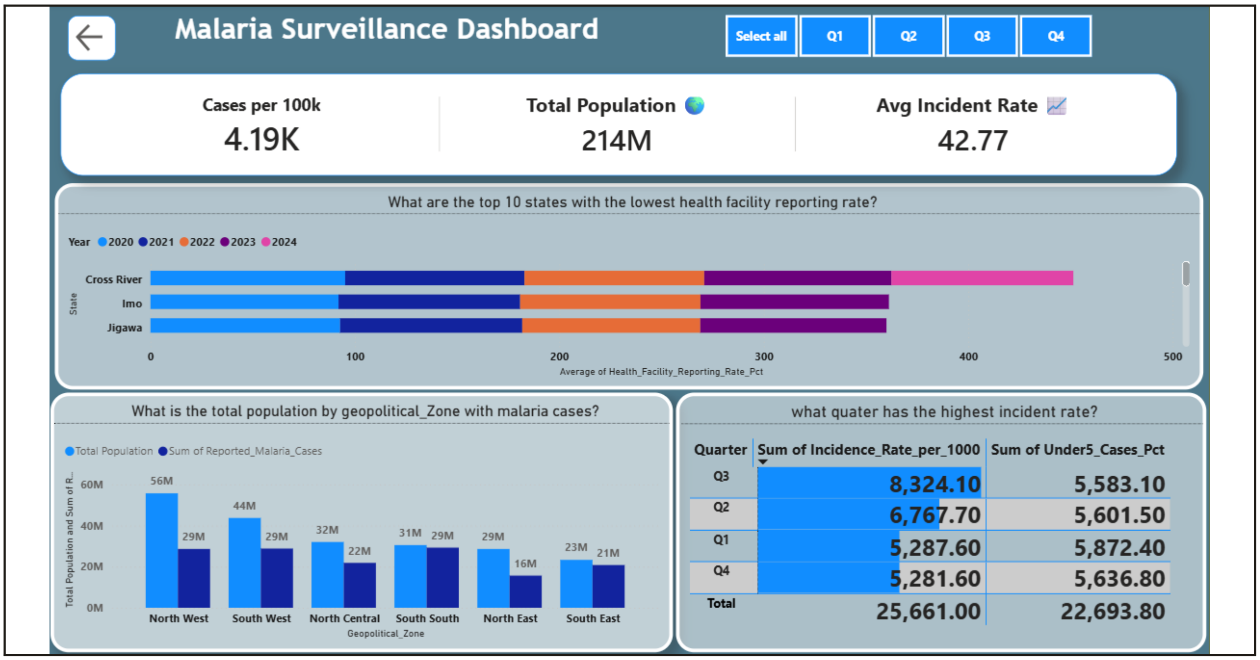

Figure 2. Geographic & Data Quality — reporting rate by state, population vs case volume by zone, and quarterly incidence table.

- Header: same title bar with a Quarter tile slicer (Select all · Q1 · Q2 · Q3 · Q4).
- KPI Row: Cases per 100k (4.19K) · Total Population 🌍 (214M) · Avg Incident Rate 📈 (42.77).
- "What are the top 10 states with the lowest health facility reporting rate?" — stacked bar by State (Cross River, Imo, Jigawa shown), segmented by Year 2020–2024.
- "What is the total population by geopolitical_Zone with malaria cases?" — clustered column, Total Population vs Sum of Reported_Malaria_Cases across all 6 zones; North West highest population (56M).
- "what quater has the highest incident rate?" — table, Quarter × Sum of Incidence_Rate_per_1000 × Sum of Under5_Cases_Pct; Q3 highest at 8,324.10.

5. Key Findings

5.1 Overall Burden Is Declining

- National incidence fell every year on record: from 45.5 per 1,000 (2020) to 38.6 per 1,000 (2023) — a 15.2% reduction.
- Total reported cases dropped from approximately 38.9M (2020) to 33.0M (2023).
- Malaria deaths declined in parallel, from roughly 105,000 (2020) to 89,000 (2023).

5.2 ITN Coverage Rose — But Its Relationship to Incidence Is Weak

- Average ITN (insecticide-treated net) coverage climbed substantially, from 50.8% (2020) to 67.7% (2023) — a 16.9 percentage-point increase.
- Despite this, the state-quarter-level statistical correlation between ITN coverage and incidence rate is only $r = -0.09$ — a very weak relationship.
- Interpretation: the national-level trend (coverage up, incidence down) does not hold as a strong direct relationship at the state level. Other factors are likely driving the incidence decline more than ITN coverage alone.

5.3 Rainfall Is a Stronger Predictor Than ITN Coverage

- Average rainfall correlates with incidence rate at $r = 0.53$ — notably stronger than the ITN relationship.
- Every year in the dataset shows the same pattern: incidence rises sharply in Q3 (July–September, the rainy season, averaging ~230mm rainfall) and falls in Q1 (dry season, ~40mm rainfall).
- Q3 average incidence (56.24 per 1,000) is roughly 68% higher than Q1 average incidence (33.52 per 1,000).

5.4 Burden Is Geographically Concentrated in the South

- South South and South East zones carry the country's heaviest average incidence (57.06 and 54.55 per 1,000 respectively) — more than double the rate seen in North Central/FCT areas.
- The highest-burden individual states are Edo, Rivers, Akwa Ibom, Delta, Abia, Bayelsa, Imo, and Anambra — all in the South South or South East zones.
- The lowest-burden states are concentrated in the North West and North East zones, alongside FCT/North Central.

5.5 Vulnerable Groups Carry Roughly Half the National Burden

- Children under 5 account for an average of 37.8% of all reported cases — the single largest at-risk group, representing an estimated 54.3 million cases over the 4-year period.
- Pregnant women account for an average of 12.0% of all reported cases — an estimated 17.1 million cases over the same period.
- Combined, these two groups represent approximately 49.8% of every reported case, nationally, in every year and every quarter studied.
- Critically, this combined share is statistically flat — it does not correlate with incidence rate ($r = 0.007$ for under-5 share) and did not shift materially even as total case counts fell 15% between 2020 and 2023. This indicates the vulnerable-group share is a stable demographic constant in this dataset, not a signal of burden severity.

5.6 Data Quality Varies by State

- Health Facility Reporting Rate averages 88.1% nationally, but a subset of states (including Cross River, Imo, and Jigawa in the current dashboard view) report consistently lower rates.
- States with lower reporting rates may have real incidence higher than what is officially recorded — this is flagged as a limitation, not a finding of lower actual burden.

5.7 States With Both High Case Volume and High Death Counts

- The Top 15 case-volume-vs-death-count scatter chart shows Lagos positioned furthest along both dimensions, followed by Rivers, Oyo, Ondo, Kogi, Niger, and Sokoto.
- This is a different lens from incidence rate: a state can have a moderate incidence rate but still contribute disproportionately to national case and death totals due to sheer population size (e.g. Lagos).
- The near-linear upward trend across all 15 states is consistent with the flat 0.27% national case fatality rate — death counts scale proportionally with case volume.

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6. Corrections Made & Recommendations

6.1 Corrections Applied During Development

Issue Identified	Root Cause	Fix Applied
Total Population showing 3bn	SUM() aggregated Estimated_Population across all 16 repeated quarterly rows per state	Changed to CALCULATE(SUM(...), filtered to one snapshot period) → now correctly shows 214M
Incidence Rate chart showing values up to 6,000	SUM() applied to a per-1,000 rate field instead of AVERAGE()	Changed aggregation to AVERAGE() → now correctly ranges 0–60
Pregnant Women / Under-5 donut chart showing "K" values instead of percentages	SUM() applied to percentage fields	Changed aggregation to AVERAGE()
DAX syntax error (missing comma) in Case Fatality Rate measure	DIVIDE() arguments not properly comma-separated	Corrected to DIVIDE(SUM(Deaths), SUM(Cases), 0) * 100
Geopolitical Zone count showing 7 instead of 6	FCT was split out as its own pseudo-zone during initial cleaning	Merged FCT into North Central via Power Query Table.ReplaceValue; confirmed 6 distinct values, 0% errors

6.2 Outstanding Items — Recommended Before Final Submission

- Fix the "lowest health facility reporting rate" chart (Page 2): it is currently a stacked bar by Year reaching 400–500 on the x-axis, which is not possible for a percentage field. Switch to a single Average per state (not stacked by year), or convert to a Clustered Bar Chart if the year breakdown is intentional.
- Cap the Year slider at 2023 (or clearly label 2024 as "partial/incomplete") so viewers cannot inadvertently include the incomplete 2024 data in filtered views.
- Consider adding a scatter chart of ITN Coverage (x-axis) vs Incidence Rate (y-axis) to visually support Finding 5.2 — this correlation is currently stated in narrative form but not yet visualized.
- Consider adding a scatter chart of Rainfall (x-axis) vs Incidence Rate (y-axis) to visually support Finding 5.3.

- Sync the Year, Geopolitical_Zone, and Quarter slicers across both pages (View → Sync Slicers) so filtering behaves consistently throughout the report.

6.3 Methodological Limitations to State in the Capstone Write-Up

- This dataset is synthetic/illustrative and was built for dashboarding and analytical practice — it is not sourced from NMEP, NHMIS/DHIS2, or NDHS real-world surveillance systems.
- 2024 data is incomplete (8 of 37 states, Q1 only) and has been excluded from all reported totals and trend calculations.
- Under-5 and pregnant-women case counts are derived estimates (reported cases × reported percentage share), not independently verified sub-group counts.
- Any trend projection beyond 2023 (if included) is a naive linear extrapolation from only 4 annual data points and should not be treated as a validated epidemiological forecast.

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7. Summary

The project followed a complete analytical workflow, in sequence: data cleaning and validation (Section 2), DAX measure development (Section 3), dashboard construction (Section 4), and the resulting findings (Section 5). The core narrative — declining but geographically concentrated malaria burden, driven more by rainfall/seasonality than by ITN coverage alone, with a stable ~50% share borne by children under 5 and pregnant women — is well supported by the cleaned data and consistently represented across both dashboard pages.

With the outstanding corrections in Section 6.2 applied, this dashboard is ready to support a capstone presentation or written submission.